Master Learning Plan v8

Medical-Imaging + Geometric/Molecular-ML — Research-Engineer Readiness — Master Learning Plan (v8)

Owner: MD Hashim · **Window:** opens now (mid-Aug 2026); understanding-gated, not calendar-gated · **Total effort:** ~1,870 focused hours · **Structure:** a job-critical spine (Blocks 1–2) + a parallel/later **QML deep track** + three continuous threads (DSA/C++, implement-by-hand ML, research output) + a **Confusion Buffer** scaffold for hard concepts + an application overlay that starts **now**.

> **What changed from v7, and why.** v7 assumed your math/ML fundamentals were *reactivation* and your quantum work was a *production strength*, so it compressed both. You've corrected the record: **math and ML/DL fundamentals are a genuine build for you** (much terminology forgotten, modeling intuition still forming), and **quantum was vendor-oversight, not hands-on depth** — and you *want* real QML expertise. You're also a **slow-but-intuitive learner** who does best with generous time and crystal-clear mental pictures. So v8 **keeps every sequencing and positioning win from v7** (early job ladder, Research-Engineer target, cut legacy CV, video pulled up next to imaging, spike protected, threads de-duplicated, capstones renumbered, PhD/publication/risk sections) and **reverses the foundation and QML compression**: it restores and *deepens* the math (Phase 1), the fundamentals (Phase 0), and the DL-theory/modeling core (Phase 3); it restores quantum to a robust **deep track** (Phase Q) — kept *off* the early-job critical path so it never blocks early value, but generous because you want the expertise; and it adds a first-class **Confusion Buffer** system — your shock absorber for complex and confusing terms. Hours are no longer minimized; the optimization is now **sequencing for early value + robustness where you're weak**. All v6/v7 URLs are preserved.

Changelog (v7 → v8) — read this first

Area	v7	v8
Learner model	Math/fundamentals = reactivation; quantum = strength	**Math/fundamentals = genuine build; quantum = genuine gap you want to close; slow-intuitive learner**
Total hours	~1,360 (minimized)	**~1,870** (robust; targeted depth where weak, still leaner than v6 because legacy CV is cut)
Phase 0 (fundamentals)	35 (quick reactivation)	**55** — thorough rebuild + terminology glossary + spoken drills
Phase 1 (math)	120 (compressed)	**205** — full intuition-first math build; linear algebra restored, extra scaffolding
Phase 3 (DL theory)	105	**160** — restored + deepened (this is exactly your "modeling fundamentals/intuition" gap)
Quantum / QML	~100, folded away	**Phase Q — ~305 (deep track), off the critical path**, dedicated capstone, + a **QML career branch** (QpiAI etc.). The molecular spike still borrows the "quantum-informed" angle
Shock absorbers	Buffer weeks (implicit)	**The Confusion Buffer** — a concrete 8-part system for hard terms (new first-class section)
Phase 2 theory depth	Compressed (110)	**Restored to 155** (applied packaging stays lean; conv/loss/uncertainty *theory* restored)
Everything v7 fixed	Sequencing, RE-target, cut legacy CV, video-up, de-dup, capstone renumber, PhD/pub/risk	**All preserved**

The program at a glance

v8 separates a **job-critical spine** (what gets you interviewing and climbing the ladder) from a **QML deep track** (what you want long-term, kept off the critical path so it never delays early value).

Job-critical spine — Block 1 + Block 2 (~1,270 hrs).

- *Block 1 — Foundations, domain & the spike (~800 hrs):* a thorough fundamentals rebuild → a full, intuition-first math build → your medical-imaging domain (applied packaging + restored theory) → imaging+video (the fallback leg) → DL theory & modeling → production systems → the **geometric/molecular-ML spike** (the differentiator for the relocation target).
- *Block 2 — Breadth & lab-interview readiness (~470 hrs + interviewing):* GenAI & agents (domain-focused) → optional CV-breadth literacy → a research + big-tech interview crescendo (DSA at RE weight, implement-by-hand ML, ML system design, research job-talk, behavioral).

QML deep track — Phase Q (~305 hrs, parallel/later). A robust, intuition-first quantum + QML build from near-zero — because you want the expertise — deliberately sequenced *off* the early-job path (run it in parallel at low weekly intensity, or as a concentrated block after you reach Tier-2 / land an initial role). It seeds a QML career branch and deepens the molecular spike's "quantum-informed" story.

Threads (parallel, from the Phase-1 self-test gate). DSA + C++ (light), implement-by-hand ML (the highest-ROI thread, and itself a concept-cementing shock absorber), research output & networking (carries your live applications and seeds a later PhD).

The Confusion Buffer. A concrete system that turns "confusing terminology" from a blocker into a managed queue — intuition-first ordering, a living glossary, spaced re-derivation, a Feynman gate, an unblock ritual, consolidation weeks, understanding-gated checkpoints, and multi-teacher triangulation for the hardest topics.

Application overlay (from now). Your *applied* medical-imaging skill already makes you eligible for the India-fallback tier — so applications start now, in parallel, while the foundations build underneath. The learning improves your interview *performance* (curing the articulation pain point), not your eligibility.

Why this order — early value without shortchanging the foundations

You named the core tension exactly: a **vanilla model config hits a ceiling**, and you can't push past baseline performance — or articulate *why* a model behaves as it does to a panel — without solid math and fundamentals. So v8 does **not** rush the foundations. Instead it protects early value three ways that don't depend on finishing the learning first:

1. **Apply from day one in parallel.** Your applied medical-imaging skill (nnU-Net/MONAI, MRI/fluorescence) is real and already hireable at Tier 0 (Qure.ai/SigTuple/GE/NVIDIA-India). Enter that funnel now; early loops pay you in feedback and calibration even before you're at peak.
2. **Sequence the *understanding* gates so each capstone unlocks a higher tier.** Capstone 1 (segmentation) → Tier 0/1; Capstone 2 (video) → Tier 1; Capstone 3 (molecular spike) → Tier 2. You climb incrementally as the foundations mature underneath.
3. **Keep QML off the critical path.** It's a genuine, generous investment — but because the relocation labs don't hire on it, it must not delay Tiers 0–2. It runs parallel/later.

And the foundations themselves are **built for how you learn**: intuition first, generous time, spaced re-derivation, and understanding-gated checkpoints — not a calendar sprint.

Pace & the critical path (understanding-gated)

Hours are not the constraint — understanding is. The self-tests and capstones are the real gates; you advance when you can *explain, derive, and implement unaided*, not when a date passes. Given you're a slow-intuitive learner, budget generously and use the Confusion Buffer's consolidation weeks liberally.

Rough elapsed time is offered only for planning (choose a pace you can sustain without burnout — a steady ~25–35 hrs/week on top of your job is far safer over a long horizon than heroic weeks):

Milestone	Focused hrs (spine)	~elapsed @25 hrs/wk	~elapsed @35 hrs/wk
Tier-0 eligible to apply	0 (applied skill exists now)	now	now
Tier-0 interview-*ready* (fundamentals + Capstone 1)	~360	~14 wks	~10 wks
Tier-1 strengthened (+ Capstone 2 video)	~455	~18 wks	~13 wks
Tier-2 RE credible (+ spike + Capstone 3)	~900	~36 wks (~8–9 mo)	~26 wks (~6 mo)
Job-critical spine complete	~1,270	~51 wks	~36 wks
QML deep track (Phase Q)	+~305 (parallel/later)	+~12 wks if concentrated	+~9 wks if concentrated

Critical path to Tier-2 RE (relocation target): P0 → P1 (full) → P3 → **P6 spike + Capstone 3** → P5 ML-system-design slice → P9 job-talk + DSA/implement-from-scratch, with **Thread M running throughout**. Quantum (Phase Q), GenAI (P7), and legacy CV (P8) are *not* on this path.

The Confusion Buffer — your shock absorber for hard concepts *(new in v8)*

Apply this to every topic that feels slippery. It turns "I keep forgetting the terminology / I'm not intuitive about this" into a managed system. This is as important as any phase — use it constantly.

1. **Intuition-first ordering (never start rigorous cold).** For each hard topic, go in this order: **(a) visual/intuitive explainer first** (3Blue1Brown, StatQuest, a good blog) to build the mental picture → **(b) the rigorous source** (MIT/Strang, a paper) to make it precise → **(c) implement or derive it by hand** to make it yours. Starting with the rigorous source before you have the picture is the #1 cause of slow-learner grind.
2. **Living glossary ("terminology war chest").** One doc. Every confusing term gets: a one-line plain-English definition, a tiny concrete example, and where it shows up. Add to it every session. This directly fixes "forget most of the terminology" — you stop re-learning terms and start *looking them up in your own words*.
3. **Spaced re-derivation (Anki-style, for concepts).** Any concept that felt hard goes into a spaced queue: re-explain/re-derive it at **1 day → 3 days → 1 week → 1 month**. This is Thread M's muscle-memory logic applied to *understanding*, not just code. Tools: Anki https://apps.ankiweb.net/ or a simple dated checklist.
4. **The Feynman gate (the real "done" test).** A concept isn't done until you can explain it, out loud, in plain language, **to an imagined 12-year-old, without notes**. If you can't, you've found the exact gap — go back to step 1 for that sub-piece. (This *is* your interview-articulation training.)
5. **The unblock ritual (no grinding).** When stuck >30 min: write the *exact* confusion in a confusion log, then **switch teacher/modality** (different lecturer, a blog, a video, a worked example) rather than re-reading the same source. Slow-intuitive learners lose the most time grinding one explanation that isn't clicking. A different angle usually unlocks it fast.
6. **Consolidation weeks (schedule-level shock absorber).** Every ~4–5 weeks: **no new material.** Only re-derive, review the glossary, and re-implement your shakiest concepts. Bank these liberally; skipping them is how confusion compounds.
7. **Understanding-gated checkpoints.** The phase self-tests are pass/fail on *explain + derive + implement unaided*. **You do not advance until you pass** — dates flex, the standard doesn't. This protects you from carrying shaky foundations into harder material.
8. **Multi-teacher triangulation for the hardest ~6 topics.** For the concepts most people (and you) find hardest — **backprop, attention & the $\sqrt{d_k}$ scaling, SVD/eigendecomposition, diffusion/score-matching, equivariance, and VQE/barren plateaus** — the plan deliberately lists **2-3 different teachers each** (below). If one doesn't click, another will. Intuition through triangulation.

Triangulation sources for the hard six:

- **Backprop:** 3B1B NN https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi · Karpathy micrograd https://github.com/karpathy/micrograd · d2l.ai backprop https://d2l.ai/
- **Attention / √dₖ:** Annotated Transformer http://nlp.seas.harvard.edu/annotated-transformer/ · 3B1B attention https://www.youtube.com/@3blue1brown · Lilian Weng https://lilianweng.github.io/
- **SVD / eigendecomposition:** 3B1B LA https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab · MIT 18.06 (Strang) https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/ · MIT 18.065 https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/
- **Diffusion / score-matching:** Yang Song blog https://yang-song.net/blog/2021/score/ · Lilian Weng diffusion https://lilianweng.github.io/posts/2021-07-11-diffusion-models/ · CS236 https://www.youtube.com/playlist?list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8
- **Equivariance:** Bronstein GDL https://geometricdeeplearning.com/lectures/ · e3nn tutorial https://blondegeek.github.io/e3nn_tutorial/ · Veličković https://www.youtube.com/results?search_query=petar+velickovic+gnn+lectures
- **VQE / barren plateaus:** PennyLane demos https://pennylane.ai/qml/demos_quantum-chemistry · Quantum Country https://quantum.country/qcvc · Schuld "Taking stock of QML" https://www.youtube.com/results?search_query=maria+schuld+taking+stock+quantum+machine+learning

---

Threads (the muscle-memory + output scaffold)

*All three start at the **Phase-1 self-test** (once you can derive backprop + attention unaided — understanding-gated, not a fixed date) and run continuously. Do them first thing, before the day's phase work. On heavy weeks or during consolidation weeks, dial the combined thread load to ~2–3 hrs — consistency beats volume.*

Thread T — DSA + C++ · ~2.5 hrs/week

RE loops need medium-hard DSA fluency (typically two coding rounds), not a FAANG-hard grind. C++ is nice-to-have (DeepMind CoderPad; edge work), not core for a JAX/PyTorch science lab — so the C++ stage is short.

Stage T1 — C++ essentials (compressed): types, references/pointers, RAII, the STL, classes, templates, smart pointers.

- learncpp.com (sections 1–17) — https://www.learncpp.com/ · The Cherno — https://www.youtube.com/@TheCherno · Reference — https://en.cppreference.com/

Stage T2 — DSA, sustained: ~2–3 problems/week, spaced, pattern-based.

- NeetCode 150 — https://neetcode.io/ · LeetCode — https://leetcode.com/ · MIT 6.006 — https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/ · Abdul Bari — https://www.youtube.com/@abdul_bari · Reference: CLRS

Checkpoint: by the crescendo, implement the core structures + standard patterns from memory in C++; a random LeetCode medium in ~30–35 min.

Thread M — Implement-by-hand ML (muscle memory) · ~2 hrs/week · highest-ROI thread + a concept-cementing shock absorber

*Directly validated: DeepMind's RE coding round asks you to implement components/losses from a blank file (e.g. "implement Focal Loss from scratch in PyTorch" — one of your own domain losses). For a slow-intuitive learner this thread does **double duty**: it builds the interview skill AND cements the concepts you're learning (implementing something is the fastest way to truly understand it). Absorbs old cross-cutting habit #3.* From a **blank file, no autocomplete, no AI**, reimplement one primitive in ~30–40 min, then revisit older ones spaced.

Rotating list: backprop through an MLP · scaled dot-product + multi-head attention · a segmentation loss (Dice/Focal/Tversky) · cross-entropy from logits · LayerNorm/BatchNorm · softmax + stable log-softmax · a sampler (top-k/nucleus/temperature) · a DDPM step · logistic regression + gradient · k-means · PCA via SVD · a tiny SGD/Adam loop · positional encodings (sinusoidal/RoPE) · a KV-cache · a message-passing GNN layer · translate one primitive to JAX.

- d2l.ai https://d2l.ai/ · labml.ai https://nn.labml.ai/ · Annotated Transformer http://nlp.seas.harvard.edu/annotated-transformer/ · micrograd https://github.com/karpathy/micrograd · nanoGPT https://github.com/karpathy/nanoGPT · target rounds https://igotanoffer.com/en/advice/google-deepmind-research-engineer-interview

Thread R — Research output, networking & applications · ~1.5 hrs/week

The Tier-2/PhD differentiator and your pipeline engine. Absorbs old habits #6 (reproduce-a-paper) and #7 (positioning doc).

- **Reproduce-a-target-paper (monthly):** publicly reproduce one paper from a target team and write it up. Code: https://paperswithcode.com/
- **Write:** 1–2 short posts/month; maintain the "why me for digital biology" narrative + two job-talk decks.
- **Publish:** aim the spike capstone at a venue (see Publication strategy) — MICCAI https://www.miccai.org/ · MIDL https://midl.io/ · MLSB https://www.mlsb.io/ · LoG https://logconference.org/ · ISMB/RECOMB https://www.iscb.org/ · arXiv https://arxiv.org/ · bioRxiv https://www.biorxiv.org/ · OpenReview https://openreview.net/
- **Applications:** run the company ladder (below); from Tier 0 (now) this thread carries your live loops.

BLOCK 1 — Foundations, domain & the spike

PHASE 0 — Rapid Brush-Up → Fundamentals Rebuild: Stats, ML & DL

~55 hrs *(v7 was 35)* · a genuine rebuild + terminology glossary + spoken command · do this first · directly cures the articulation pain point

Goal: rebuild crisp, *out-loud* command of the fundamentals and the vocabulary. This is not a quick reactivation for you — much terminology has faded — so go thoroughly, intuition-first, and **build your living glossary as you go** (Confusion Buffer #2). The deliverable is *fluent spoken articulation* + a glossary you trust.

Primary resource — StatQuest: https://www.youtube.com/@statquest/playlists · https://statquest.org/

- **0.1 Statistics fundamentals (~14 hrs)** — distributions; mean/variance/SD; populations vs samples; CLT; sampling; hypothesis testing & p-values; CIs; R^2; covariance & correlation; MLE; bootstrapping; power. *Verbal checkpoint:* p-value, CI, CLT, MLE each in two sentences, no notes.
- **0.2 Classic ML (~20 hrs)** — bias–variance; train/val/test & CV; confusion matrix, sensitivity/specificity, precision/recall, ROC & AUC (and when PR is honest); linear & logistic regression; ridge vs lasso and why L1→sparsity; trees/RF/AdaBoost/GBM/XGBoost; SVMs & the kernel trick; k-means & hierarchical clustering; PCA; naive Bayes; entropy & mutual info. *Verbal checkpoint, cold:* gradient boosting vs random forest; why L1 zeros weights and L2 shrinks (at the gradient level); what AUC measures and when PR-AUC is more honest.
- **0.3 Deep-learning fundamentals (~14 hrs)** — StatQuest NN + 3Blue1Brown NN https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi : perceptron/MLP; activations; backprop; gradient descent; losses; tensors; CNN & RNN/LSTM basics; overfitting, dropout, batch/layer norm, vanishing/exploding gradients; transfer learning.
- **0.4 Deliverable — panel articulation cheat-sheet + glossary v1 (~7 hrs)** — one or two crisp spoken sentences per concept; seed the terminology war chest.

Phase 0 self-test (Feynman gate): explain 15 of the above out loud, plain language, 2–3 sentences each, no notes — as if to a panel *and* to a 12-year-old.

PHASE 1 — Foundations: Mathematics + Neural Networks from First Principles

~205 hrs *(v7 was 120 — restored and deepened)* · the bedrock · this is a genuine build, so do NOT rush it · intuition-first throughout

Goal: real mathematical maturity — the thing that lets you push a model past its vanilla ceiling and explain *why* it works. Because math is a build for you (not a refresh), this is generous and intuition-first: **watch the visual explainer, then the rigorous lecture, then derive by hand** (Confusion Buffer #1), with spaced re-derivation of anything slippery. By the end you can re-derive core results on paper and understand attention mechanically (Karpathy's GPT build), which unlocks Phase 2 and seeds Thread M.

- **1.1 Linear algebra (~72 hrs) — highest-leverage topic in the plan · restored to full.**
  - 3B1B Essence of Linear Algebra (intuition first) — https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab
  - MIT 18.06 (Strang) — the rigorous course — https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/video_galleries/video-lectures/
  - MIT 18.065 Matrix Methods (Strang) — SVD, PCA, low-rank, GD through a linear-algebra lens — https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/
  - Reference: Matrix Cookbook — https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf
  - *Whiteboard (spaced):* derive SVD from eigendecomposition; rank-r approximation geometrically; A^TA is PSD; backprop through a linear layer with matrix calculus.
- **1.2 Calculus + matrix calculus (~26 hrs).**
  - 3B1B Essence of Calculus — https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr
  - Matrix Calculus for Deep Learning (Parr & Howard) — https://explained.ai/matrix-calculus/
- **1.3 Probability & statistics (~50 hrs).**
  - Harvard Stat 110 (Blitzstein) — https://projects.iq.harvard.edu/stat110/youtube
  - Alt: MIT RES.6-012 (Tsitsiklis) — https://ocw.mit.edu/courses/res-6-012-introduction-to-probability-spring-2018/
  - *Whiteboard:* MLE for a Gaussian; conditional expectation as projection; Bayes on a non-trivial example.
- **1.4 Neural networks from first principles (~57 hrs) — the bridge · PROTECT.** Do all 10 videos with exercises, re-implementing each without looking at his code. The substrate for Thread M and the single best "make the math concrete" step in the plan.
  - Karpathy — Neural Networks: Zero to Hero — https://www.youtube.com/playlist?list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ · hub https://karpathy.ai/zero-to-hero.html

Phase 1 self-test (understanding gate → Threads T, M, R start): hand-derive backprop for a 2-layer MLP; implement micrograd-style autodiff from memory; explain why BatchNorm helps and what breaks without it; whiteboard scaled dot-product attention and derive the $\sqrt{d_k}$ scaling. Pass this before advancing.

PHASE 2 — Applied Computer Vision & Medical Image Segmentation

~155 hrs *(v7 was 110 — theory restored)* · applied packaging stays lean; the *theory/math* behind your domain is restored, because that's where your ceiling is · ends with Capstone 1 → Tier 0/1

Goal: turn what you already *do* (nnU-Net/MONAI) into what you can *explain and improve* — the conv math, the loss math, the "why it works" — plus the public portfolio piece. **Apply to Tier-0 India roles in parallel with this phase.**

- **2.1 Imaging physics for interviews (~20 hrs).**
  - Allen Elster — Q&A in MRI — https://mriquestions.com/index.html · OHBM — https://www.youtube.com/@ohbmonline/playlists · Callaghan (first 4) — https://www.youtube.com/playlist?list=PLqEMVgX2VgZcG_xR1QRdsuc-X_zEgMx-Z · iBiology fluorescence — https://www.youtube.com/playlist?list=PLF513KEDjY9YBNhwMfX6jMl3PlpdW9TMP
  - *Deliverable:* 1-page note — T1/T2/FLAIR/T1CE + fluorescence artifacts.
- **2.2 CNNs + convolution math + U-Net family + losses (~45 hrs) — restored; the math you need to break the ceiling.**
  - CS231n 2017 (Johnson) — https://www.youtube.com/playlist?list=PL3FW7Lu3i5JvHM8ljYj-zLfQRF3EO8sYv · https://cs231n.github.io/
  - Convolution arithmetic — https://arxiv.org/abs/1603.07285 · https://github.com/vdumoulin/conv_arithmetic
  - U-Net https://arxiv.org/abs/1505.04597 · V-Net https://arxiv.org/abs/1606.04797 · Attention U-Net https://arxiv.org/abs/1804.03999 · nnU-Net https://www.nature.com/articles/s41592-020-01008-z
  - Losses — survey https://arxiv.org/abs/2006.14822 · Generalized Dice https://arxiv.org/abs/1707.03237 · Focal Tversky https://arxiv.org/abs/1

810.07842 · Boundary loss https://arxiv.org/abs/1812.07032 (implement each from scratch — Thread M).
    - *Whiteboard:* U-Net with tensor shapes; why skip connections work; Dice gradient instability on empty masks; nnU-Net's "no new architecture" thesis.
- **2.3 nnU-Net + MONAI packaging (~18 hrs) — keep lean; this is your applied strength.**
    - MONAI — https://github.com/Project-MONAI/tutorials · videos https://www.youtube.com/@projectmonai/videos · nnU-Net v2 — https://github.com/MIC-DKFZ/nnUNet · MONAI integration https://github.com/Project-MONAI/tutorials/blob/main/nnunet/README.md · Kitware https://www.kitware.com/developing-custom-3d-medical-image-segmentation-solutions-using-out-of-the-box-pipelines-in-monai/
    - DigitalSreeni — https://www.youtube.com/@DigitalSreeni/playlists · code https://github.com/bnsreenu/python_for_microscopists · U-Net playlist https://www.youtube.com/playlist?list=PLZsOBAyNTZwbR08R959iCvYT3qzhxvGOE · AI for Medical Diagnosis — https://www.coursera.org/learn/ai-for-medical-diagnosis
- **2.4 Transformer-based segmentation + foundation models (~20 hrs).**
    - ViT https://arxiv.org/abs/2010.11929 · TransUNet https://arxiv.org/abs/2102.04306 · Swin-UNet https://arxiv.org/abs/2105.05537 · UNETR https://arxiv.org/abs/2103.10504 · Swin UNETR https://arxiv.org/abs/2201.01266 · SAM https://arxiv.org/abs/2304.02643 · MedSAM https://www.nature.com/articles/s41467-024-44824-z · MedNeXt https://arxiv.org/abs/2303.09975
- **2.5 3D, uncertainty, evaluation (~32 hrs) — restored.**
    - TorchIO — https://torchio.readthedocs.io/ · playlist https://www.youtube.com/playlist?list=PLfXIfDgnRPlflEHFx2N6mlbn-MGFOPiX2
    - Uncertainty (Yarin Gal) — https://www.youtube.com/results?search_query=yarin+gal+uncertainty+deep+learning
    - Metrics Reloaded — https://www.nature.com/articles/s41592-023-02151-z · Domain shift (Stanford AIMI) — https://aimi.stanford.edu/education/educational-resources
    - *3D-conv exercise:* input (D=64,H=128,W=128,C=4), kernel (3,3,3), stride (1,2,2), padding (1,1,1), 32 out-channels — output shape + param count by hand; repeat for transposed + dilated.

PHASE 2 — CAPSTONE 1 (~20 hrs) · public data, COI-safe · the Tier-0/1 gate. Notebook + 3-page write-up: preprocessing (physics-justified), architecture ablation (3 architectures), loss-function ablation, uncertainty + failure-mode analysis. Implement the losses + one architecture block from scratch (Thread M).

- **Datasets (pick one):** Medical Decathlon http://medicaldecathlon.com/ · BraTS https://www.synapse.org/brats · TCIA https://www.cancerimagingarchive.net/ · IXI https://brain-development.org/ixi-dataset/ — **or microscopy-native:** LIVECell https://github.com/sartorius-research/LIVECell · Sartorius (Kaggle) https://www.kaggle.com/competitions/sartorius-cell-instance-segmentation · BBBC https://bbbc.broadinstitute.org/
- **Angle:** portfolio + a MIDL/MICCAI-workshop-grade short paper (Thread R).

Phase 2 self-test: whiteboard U-Net forward pass + skips with shapes; why nnU-Net usually beats fancier architectures (and when not); diagnose bias-field artifacts, class imbalance, label noise, domain shift on a new dataset.

PHASE 2B — Imaging & Video (life-sciences-native)

~95 hrs · the fallback leg made modern · ends with Capstone 2 → Tier 1

Goal: extend your imaging leg into **video/temporal** — the closest genuine expansion of your strength, unlocking phenomics/cell-imaging roles (Recursion) and strengthening the India case.

- **2B.1 Video & temporal analytics (~40 hrs).**
    - VideoMAE https://arxiv.org/abs/2203.12602 · TimeSformer https://arxiv.org/abs/2102.05095 · ViViT https://arxiv.org/abs/2103.15691 · SlowFast https://arxiv.org/abs/1812.03982
    - SAM 2 https://arxiv.org/abs/2408.00714 · https://github.com/facebookresearch/sam2 · https://ai.meta.com/sam2
    - RAFT https://arxiv.org/abs/2003.12039 · https://github.com/princeton-vl/RAFT
    - ByteTrack https://arxiv.org/abs/2110.06864 · https://github.com/ifzhang/ByteTrack · OC-SORT https://arxiv.org/abs/2203.14360 · https://github.com/noahcao/OC_SORT · BoT-SORT https://arxiv.org/abs/2206.14651 · https://github.com/NirAharon/BoT-SORT
    - HOTA https://arxiv.org/abs/2009.07736 · MOTChallenge https://motchallenge.net/ · DAVIS https://davischallenge.org/ · YouTube-VOS https://youtube-vos.org/
- **2B.2 Life-sciences-native video (~30 hrs) — the imaging-leg differentiator.**
    - DeepLabCut https://www.deeplabcut.org/ · https://github.com/DeepLabCut/DeepLabCut · Nature Neuroscience https://www.nature.com/articles/s41593-018-0209-y · SLEAP https://sleap.ai/ · https://github.com/talmolab/sleap · Nature Methods https://www.nature.com/articles/s41592-022-01426-1
    - Cellpose https://www.cellpose.org/ · https://github.com/MouseLand/cellpose · TrackMate https://imagej.net/plugins/trackmate/ · Ultrack https://github.com/royerlab/ultrack · Cell Tracking Challenge http://celltrackingchallenge.net/
- **2B.3 Detection/instance breadth (~25 hrs) — feeds cell-instance work.**
    - Michigan EECS 498 (Justin Johnson) — detection/instance/panoptic — https://web.eecs.umich.edu/~justincj/teaching/eecs498/ · playlist https://www.youtube.com/playlist?list=PL5-TkQAfAZFbzxjBHtzdVCWE0Zbhomg7r

PHASE 2B — CAPSTONE 2 (video, ~included) · public data, COI-safe · Tier 1. DeepLabCut/SLEAP pose → temporal model → behavior classification; **or** Cellpose seg → TrackMate/Ultrack tracking + lineage; SAM 2 for mask propagation.

- **Datasets:** CalMS21 https://data.caltech.edu/records/s0vdx-0k302 · MABe https://www.aicrowd.com/challenges/multi-agent-behavior-challenge-2022 · Cell Tracking Challenge http://celltrackingchallenge.net/

Phase 2B self-test: SAM 2's memory mechanism; tracking-by-detection vs joint detection-and-tracking; when a video transformer beats frame-wise + tracking; cell-lineage failure modes.

PHASE 3 — Deep Learning Theory & Modern Architectures

~160 hrs *(v7 was 105 — restored and deepened)* · the modeling-fundamentals core · this is exactly your "not intuitive about modeling" gap, so go thoroughly, intuition-first

Goal: build real modeling intuition — optimization dynamics, information theory, modern architectures, generative/diffusion theory. This is where "why does my model behave this way, and how do I make it better" is answered. Triangulate the hard topics (Confusion Buffer #8).

- **3.1 Optimization for deep learning (~30 hrs).**
  - Sebastian Ruder — https://www.ruder.io/optimizing-gradient-descent/ · Boyd EE364A (targeted) — https://www.youtube.com/playlist?list=PL3940DD956CDF0622 · book https://web.stanford.edu/~boyd/cvxbook/
- **3.2 Information theory for ML (~20 hrs).**
  - David MacKay ITILA, ch 1–6 — https://www.inference.org.uk/itila/book.html
  - *Whiteboard:* cross-entropy from KL; NLL = MLE; the ELBO; mutual information in InfoNCE.
- **3.3 Transformers in depth + modern variants (~42 hrs).**
  - CS25 — https://web.stanford.edu/class/cs25/ · playlist https://www.youtube.com/playlist?list=PLoROMvodv4rNiJRchCzutFw5ItR_Z27CM · Lilian Weng — https://lilianweng.github.io/ · Annotated Transformer — http://nlp.seas.harvard.edu/annotated-transformer/
  - RoPE https://arxiv.org/abs/2104.09864 · RMSNorm https://arxiv.org/abs/1910.07467 · FlashAttention https://arxiv.org/abs/2205.14135 · GQA/MQA https://arxiv.org/abs/2305.13245 · SwiGLU https://arxiv.org/abs/2002.05202 · MoE/Switch https://arxiv.org/abs/2101.03961
- **3.4 Vision transformers + modern ConvNets (~26 hrs).**
  - ViT lectures (Lucas Beyer) — https://www.youtube.com/results?search_query=lucas+beyer+vision+transformer · ConvNeXt https://arxiv.org/abs/2201.03545 · DINOv2 https://arxiv.org/abs/2304.07193 · MAE https://arxiv.org/abs/2111.06377 · HF CV course — https://huggingface.co/learn/computer-vision-course
- **3.5 Generative modeling & diffusion (~42 hrs) · PROTECT — feeds the spike & the relocation target's generative lane.**
  - CS236 (Ermon) — https://www.youtube.com/playlist?list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8 · DDPM https://arxiv.org/abs/2006.11239 · Yang Song score-based https://yang-song.net/blog/2021/score/ · Lilian Weng diffusion https://lilianweng.github.io/posts/2021-07-11-diffusion-models/ · HF Diffusion (units 1–2) — https://huggingface.co/learn/diffusion-course

Phase 3 self-test: derive the diffusion loss (variational + score-matching views); compare positional encodings; explain self-supervised pretraining in low-data regimes; explain what actually changes training dynamics when you add momentum / warmup / a schedule.

PHASE 5 — Full-Stack / Production AI Systems

~140 hrs *(v7 was 105 — some restored)* · production breadth + ML-systems-design (in the RE loop)

Goal: upgrade from "I built it" to "I can defend every architectural choice and design the next-gen version." (Numbered 5 to preserve v6/v7 alignment; runs after the spike-prep or in parallel as capacity allows.)

- **5.1 LLM internals (~28 hrs).** CS336 (targeted) — https://stanford-cs336.github.io/spring2024/ · playlist https://www.youtube.com/playlist?list=PLoROMvodv4rOY23Y0BoGoBGgQ1zmU_MT_ · HF NLP — https://huggingface.co/learn/nlp-course
- **5.2 RAG at senior level (~24 hrs).** Hamel Husain — https://www.oreilly.com/radar/what-we-learned-from-a-year-of-building-with-llms-part-i/ · Eugene Yan patterns — https://eugeneyan.com/writing/llm-patterns/ · Anthropic contextual retrieval — https://www.anthropic.com/news/contextual-retrieval · LlamaIndex https://docs.llamaindex.ai/ · RAGAS https://docs.ragas.io/ · DSPy https://dspy.ai/
- **5.3 MLOps + ML systems design (~40 hrs) · KEEP.** Chip Huyen — https://huyenchip.com/ml-interviews-book/contents/8.1.1-ml-systems-design.html · CS329S — https://stanford-cs329s.github.io/ · Made With ML — https://madewithml.com/ · Full Stack DL — https://fullstackdeeplearning.com/course/2022/
- **5.4 Distributed training + inference optimization (~24 hrs).** Raschka — https://magazine.sebastianraschka.com/ · HF multi-GPU — https://huggingface.co/docs/transformers/en/perf_train_gpu_many · vLLM https://arxiv.org/abs/2309.06180 · quantization (GPTQ/AWQ/GGUF/FP8) · ZeRO https://arxiv.org/abs/1910.02054
- **5.5 Eval / observability / safety + ML-systems-design prep (~24 hrs).** Anthropic Constitutional AI — https://www.anthropic.com/news/constitutional-ai · Eugene Yan evals — https://eugeneyan.com/writing/llm-evaluators/ · observability (LangSmith / Phoenix / Helicone) on a real RAG system; *ML*-systems-design whiteboards (RAG for HCPs; brain-seg pipeline with domain-shift handling; retraining + monitoring for a drug-discovery digital twin).

PHASE 6 — Geometric / Molecular ML — THE SPIKE

~185 hrs *(v7 was 175)* · the differentiator for the relocation target + the PhD seed · a genuine gap, kept robust · ends with Capstone 3 (flagship) → Tier 2

Goal: close the geometric-deep-learning / molecular-AI gap that connects all your work — the single highest-reward phase for the relocation target. Your growing QML understanding (Phase Q) makes the "quantum-informed" angle here uniquely credible, on 100% public benchmarks.

- **6.1 Geometric deep learning + graph ML (~48 hrs).** CS224W core (message passing, GraphSAGE, GAT, expressivity) — https://web.stanford.edu/class/cs224w/ · 2023 https://snap.stanford.edu/class/cs224w-2023/ · playlist https://www.youtube.com/playlist?list=PLoROMvodv4rPLKxIpqhjhPgdQy7imNkDn · Bronstein GDL — https://geometricdeeplearning.com/lectures/ · proto-book https://arxiv.org/abs/2104.13478 · Veličković https://www.youtube.com/results?search_query=petar+velickovic+gnn+lectures · PyG https://pytorch-geometric.readthedocs.io/ · DGL https://www.dgl.ai/
- **6.2 Molecular ML + equivariant networks (~55 hrs) · the core.** DeepChem https://deepchem.io/ · tutorials https://deepchem.io/tutorials/ · TDC https://tdcommons.ai/ · SchNet https://arxiv.org/abs/1706.08566 · DimeNet++ https://arxiv.org/abs/2011.14115 · GemNet https://arxiv.org/abs/2106.08903 · MolFormer https://arxiv.org/abs/2106.09553 · Cohen & Welling https://arxiv.org/abs/1602.07576 · **Equivariant stack:** EGNN https://arxiv.org/abs/2102.09844 · e3nn https://e3nn.org/ · https://github.com/e3nn/e3nn · tutorial https://blondegeek.github.io/e3nn_tutorial/ · NequIP https://github.com/mir-group/nequip · paper https://www.nature.com/articles/s41467-022-29939-5 · MACE https://github.com/ACEsuit/mace · paper https://arxiv.org/abs/2206.07697 · Allegro https://github.com/mir-group/allegro · Equiformer https://arxiv.org/abs/2206.11990 · curated list https://github.com/Eipgen/Neural-Network-Models-for-Chemistry
- **6.3 Protein structure + equivariance (~27 hrs).** AlphaFold 2 — AlQuraishi explainer https://www.youtube.com/results?search_query=AlphaFold+2+explained+AlQuraishi · paper https://www.nature.com/articles/s41586-021-03819-2 · ESM/ESMFold https://github.com/facebookresearch/esm · ESM Atlas https://esmatlas.com/ · Boltz https://github.com/jwohlwend/boltz · OpenFold https://github.com/aqlaboratory/openfold
- **6.4 JAX + mock interviews + portfolio (~22 hrs).** Recorded, reviewed: 45-min segmentation deep-dive; 30-min behavioral (STAR, 4 stories). **Translate one capstone component to JAX** — https://jax.readthedocs.io/ · Equinox https://docs.kidger.site/equinox/ . Paper-of-the-

week + write-ups (Thread R).

PHASE 6 — CAPSTONE 3 (~33 hrs, flagship) · public quantum-chemistry data, COI-safe · unlocks Tier 2 · the Isomorphic-differentiator piece. Train an **E(3)-equivariant GNN** (SchNet → DimeNet++ → EGNN/NequIP-style) to predict molecular properties; ablate equivariance; **fold in the quantum-informed angle** (delta-ML toward CCSD; the DFT-vs-quantum-computing data-generation story — connects to Phase Q). Implement the message-passing core from scratch (Thread M); translate one component to JAX.

- **Datasets:** QM9 https://quantum-machine.org/datasets/ · PCQM4Mv2 https://ogb.stanford.edu/kddcup2021/pcqm4m/ · MoleculeNet https://moleculenet.org/ · OGB https://ogb.stanford.edu/ · TDC https://tdcommons.ai/
- **Angle:** portfolio + an MLSB / AI4Science / LoG workshop paper (Thread R). **This is the flagship — do it well.**

Phase 6 self-test: message passing vs convolution; what equivariance buys you and why it matters for molecules; when a GNN beats a fingerprint baseline; how AlphaFold's structure module uses equivariance; the quantum-chemistry data-generation story end-to-end.

PHASE Q — Quantum Computing & QML (the deep track) *(restored & expanded in v8)*

~305 hrs · a genuine build from near-zero, because you want the expertise · OFF the early-job critical path · intuition-first, generously paced

*You clarified your quantum exposure was overseeing a vendor (QpiAI's retrosynthesis work), not hands-on depth — and you want to actually become good at QML. So this is a real, thorough build, not a refresh. It's deliberately **off the critical path**: the life-sciences relocation labs don't hire on quantum, so Phase Q must never delay Tiers 0–2. Two ways to run it: **(A)** in parallel at low intensity (a few hrs/week over a long horizon, as a fourth "thread"), or **(B) — recommended for a slow-intuitive learner —** as a **concentrated block after you reach Tier-2 / land an initial role**, so you can give it the deep focus that makes it click. Pull it earlier only if you target a QML-specific role (see the QML career branch). It also deepens the molecular spike's "quantum-informed" story and seeds a QML career.*

Intuition-first entry (do these before the rigorous courses — Confusion Buffer #1):

- Quantum Country — https://quantum.country/qcvc (spaced-repetition primer — ideal for you)
- MIT 8.04 (Adams) — the physics picture — https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/video_galleries/lecture-videos/
- **Q.1 Group theory & symmetry (~22 hrs) — shared with the spike's equivariance.**
  - Group theory for physicists — https://www.youtube.com/results?search_query=group+theory+for+physicists+lectures · GDL lectures (Lie-group/equivariance) — https://geometricdeeplearning.com/lectures/
- **Q.2 QM math: Hilbert spaces, operators, tensor products (~25 hrs).**
  - MIT 8.04 https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/ · Quantum Country https://quantum.country/qcvc · (optional) Schuller — https://www.youtube.com/playlist?list=PLPH7f_7ZlzxTi6kS4vCmv4ZKm9u8g5yic
- **Q.3 Quantum computing fundamentals — Watrous / IBM track (~80 hrs) · the rigorous core.**
  - Basics of Quantum Information — https://learning.quantum.ibm.com/course/basics-of-quantum-information · YouTube https://www.youtube.com/playlist?list=PLOFEBzvs-VvrXTMy5Y2IqmSaUjfnhvBHR
  - Fundamentals of Quantum Algorithms — https://learning.quantum.ibm.com/course/fundamentals-of-quantum-algorithms
  - General Formulation of Quantum Information — https://learning.quantum.ibm.com/course/general-formulation-of-quantum-information
  - Foundations of Quantum Error Correction — https://learning.quantum.ibm.com/course/foundations-of-quantum-error-correction
  - Companion: Nielsen & Chuang, ch 1–4, 8, 10
- **Q.4 Variational quantum algorithms — VQE, ADAPT-VQE, DMET, QAOA (~45 hrs).**
  - PennyLane Codebook — https://pennylane.ai/codebook · quantum-chemistry demos — https://pennylane.ai/qml/demos_quantum-chemistry
  - VQE https://www.nature.com/articles/ncomms5213 · ADAPT-VQE https://www.nature.com/articles/s41467-019-10988-2 · DMET-VQE https://arxiv.org/abs/2108.08987 · QAOA https://arxiv.org/abs/1411.4028 · Hadfield https://arxiv.org/abs/1709.03489 · Pauli grouping https://arxiv.org/abs/1907.13117
- **Q.5 Quantum machine learning theory (~42 hrs) — the honest-take material.**
  - PennyLane QML topic (Schuld-curated) — https://pennylane.ai/topics/quantum-machine-learning · Maria Schuld "Taking stock of QML" — https://www.youtube.com/results?search_query=maria+schuld+taking+stock+quantum+machine+learning · Schuld & Petruccione, ch 4–7
  - QML=kernels https://arxiv.org/abs/2101.11020 · Barren plateaus https://www.nature.com/articles/s41467-018-07090-4 · Expressibility https://arxiv.org/abs/1905.10876 · Advantage-goal https://arxiv.org/abs/2203.01340 · PQC encodings https://arxiv.org/abs/2008.08605
  - Courses to pull from: Péré https://github.com/Christophe-pere/QML-Course · Hjorth-Jensen https://github.com/CompPhysics/QuantumComputingMachineLearning
- **Q.6 Quantum chemistry on quantum computers + retrosynthesis context (~60 hrs).** Ties directly to what you oversaw at QpiAI.
  - Qiskit Global Summer School — https://www.youtube.com/@qiskit/playlists · IBM Quantum Learning + Sample-based/Krylov Quantum Diagonalization — https://learning.quantum.ibm.com/ · https://www.ibm.com/quantum/blog/iql-migration

PHASE Q — CAPSTONE Q (~30 hrs) · public molecules, COI-safe. Take a **public** molecule (H_2/LiH/H_2O or a QM9 subset — **not** Lilly molecules). Re-implement three ways — vanilla VQE + UCCSD, ADAPT-VQE, DMET-VQE — benchmark against classical CCSD; a write-up defending every choice (ansatz, optimizer, measurement grouping, noise mitigation, basis set) with a barren-plateau analysis. Optionally add a classical-ML surrogate (delta-ML toward CCSD) — the "quantum × ML" story that connects to Capstone 3.

Phase Q self-test: derive VQE from the variational principle; the parameter-shift rule and why it works; the barren-plateau argument; VQE vs

QAOA — when each and when *no* quantum algorithm; an honest take on QML hype vs promise (cite Schuld); DMET fragmentation and the classical-quantum interface. *(Feynman gate: explain each to a smart non-physicist.)*

BLOCK 2 — Breadth & Research/Lab-Interview Readiness

~470 hrs + interviewing · Threads T + M + R continue underneath

PHASE 7 — Generative AI & Agentic AI (domain-focused)

~110 hrs *(v7 was 95)* · competent, not specialist · off the Tier-0/Tier-2 critical path — run lighter/later · ends with optional Capstone 4

Goal: *competent* design/adaptation/orchestration of modern GenAI/agentic systems at a domain-focused level.

- **7.1 Advanced LLMs — adaptation & inference (~32 hrs).** LoRA https://arxiv.org/abs/2106.09685 · QLoRA https://arxiv.org/abs/2305.14314 · Raschka LoRA https://magazine.sebastianraschka.com/ · InstructGPT/RLHF https://arxiv.org/abs/2203.02155 · DPO https://arxiv.org/abs/2305.18290 · HF TRL · HF LLM course https://huggingface.co/learn/llm-course · Chip Huyen *AI Engineering* (O'Reilly, 2025)
- **7.2 Multimodal & vision-language models (~24 hrs).** CLIP https://arxiv.org/abs/2103.00020 · LLaVA https://arxiv.org/abs/2304.08485 · Flamingo https://arxiv.org/abs/2204.14198
- **7.3 Agents & agentic systems (~35 hrs).** HF Agents Course https://huggingface.co/learn/agents-course · HF MCP Course https://huggingface.co/learn/mcp-course · Anthropic Building Effective Agents https://www.anthropic.com/research/building-effective-agents · Andrew Ng patterns https://www.deeplearning.ai/short-courses/ · LangGraph https://langchain-ai.github.io/langgraph/ · patterns: ReAct, tool/function calling, reflection, planning, memory, multi-agent, agent eval.

PHASE 7 — CAPSTONE 4 (~19 hrs, optional) · public literature, COI-safe. A multi-step research/analysis agent (RAG + tool use + memory + eval) over **open** scientific literature. *Optional — not on the critical path.*

- **Corpora/tools:** PMC-OA https://www.ncbi.nlm.nih.gov/pmc/tools/openftlist/ · arXiv https://arxiv.org/ · PubMed E-utilities https://www.ncbi.nlm.nih.gov/books/NBK25501/ · Semantic Scholar API https://www.semanticscholar.org/product/api. **Never ingest Lilly internal docs/data.**

Phase 7 self-test: explain LoRA and why it's parameter-efficient; RLHF vs DPO; when an agent beats a single well-prompted call; walk the ReAct loop; how you'd evaluate agent reliability.

PHASE 8 — CV-breadth literacy (classical / 3D / edge) — *optional appendix*

~30 hrs · literacy only · NOT expertise, NOT positioned as seniority

Goal: enough literacy to not freeze on a classical-vision question — not a CV-expert build. The useful video/temporal material is in Phase 2B; the heavy classical/3D/SLAM/NeRF/edge-C++ content is de-prioritized (legacy from the abandoned "generic senior CV" goal). Do only with slack; never claim CV seniority.

- **8.1 Classical & geometric CV — literacy (~10 hrs).** FPCV (Nayar) — skim — https://fpcv.cs.columbia.edu/ · Coursera https://www.coursera.org/specializations/firstprinciplesofcomputervision · Stachniss https://www.youtube.com/@CyrillStachniss · Reference: Hartley & Zisserman (as needed)
- **8.2 Deployment/edge literacy (~15 hrs).** ONNX https://onnx.ai/ · TensorRT https://developer.nvidia.com/tensorrt · OpenCV https://docs.opencv.org/ · PyImageSearch https://pyimagesearch.com/ · CUDA basics (NVIDIA intro)
- **8.3 3D vision / neural rendering — awareness (~5 hrs).** NeRF https://arxiv.org/abs/2003.08934 · 3D Gaussian Splatting https://arxiv.org/abs/2308.04079 · *deploy datasets:* KITTI https://www.cvlibs.net/datasets/kitti/ · TUM RGB-D https://cvg.cit.tum.de/data/datasets/rgbd-dataset · ETH3D https://www.eth3d.net/

Phase 8 self-test (literacy bar): pinhole camera model + epipolar constraint at a whiteboard; what TensorRT does; when 3D methods matter — enough to converse, not to claim expertise.

PHASE 9 — Research + Big-Tech Interview Crescendo

~125 hrs *(v7 was 115)* · peak the interview-specific skills right as you interview · recalibrated to the RE loop · unlocks Tier 3

Goal: convert the year-long threads into performance. RE loop = **2 medium-hard coding rounds + ML fundamentals + ML system design + behavioral**, plus a research discussion for research-adjacent roles.

- **9.1 DSA — RE weight (~40 hrs) — crescendo of Thread T.** Timed NeetCode 150 → Blind 75 → company-tagged — https://neetcode.io/ · https://leetcode.com/ ; patterns under pressure — DesignGurus https://www.designgurus.io/ . Out loud, blank editor, ~35 min, **code must run** (CoderPad-style). *Medium-hard fluency, not a hard-grind.*
- **9.2 Implement-by-hand ML (~35 hrs) — crescendo of Thread M · highest value.** Timed, unaided, from a blank file: attention, a loss (Dice/Focal), a sampler, a diffusion step, a training loop, backprop, a message-passing layer. Plus an ML math/theory sweep (gradient computation, KL properties, sampling, convergence intuition, the "L1 sparsity via gradients" mechanism — asked verbatim).
- **9.3 ML system design (~25 hrs) — distinct from Phase 5.** Focus on *ML* system design (training/serving/data/eval/monitoring, domain shift), not generic distributed systems. ByteByteGo — https://bytebytego.com/ · https://www.youtube.com/@ByteByteGo · Gaurav Sen https://www.youtube.com/@gkcs . *(DDIA / MIT 6.824 https://pdos.csail.mit.edu/6.824/ optional/de-emphasized.)*
- **9.4 Research job-talk + behavioral (~25 hrs) · KEEP.** A 20-min talk + Q&A on each of your two strongest capstones (segmentation/video + the molecular spike). ~6–8 STAR stories mapped to each org's signals; low-level/OO design only for loops that include it; live peer mocks — Pramp / interviewing.io. **Research loops ban AI tools — rehearse fully unaided.**

Phase 9 self-test: a random LeetCode medium-hard in ~40 min with clean, running code; implement attention + a loss + a sampler + a GNN layer from a blank file, unaided, timed; a full *ML* system-design whiteboard in 45 min; a crisp 20-min job-talk; four STAR stories from memory. **→ Tier 3.**

(Active interviewing: keep DSA, implement-by-hand ML, and ML system design warm with ~4–5 hrs/week maintenance; use the buffer for on-sites, take-homes, and rest.)

Application overlay & company target ladder

*Your applied medical-imaging skill makes you eligible at Tier 0 **now** — enter the funnel and climb as each capstone lands. Match the tier; early loops pay you in feedback. "Relocate" flags roles outside India.*

Tier	Unlocked by	Apply to
Tier 0	Applied skill (now) + Capstone 1 strengthens	India medical-imaging AI + accessible; stretch Google-India
Tier 1	Capstone 1 + Capstone 2 (imaging + video)	broader medical/biomedical imaging + video; stronger Google-India case
Tier 2	Capstone 3 (molecular GNN, flagship)	AI drug-discovery labs; **Isomorphic / DeepMind-science RE realistic**
Tier 3	Interview crescendo (P9)	full top-lab loops
QML branch	Capstone Q (Phase Q)	quantum-for-science roles (see below) — pull earlier only if this is a target

Tier 0 — India-domestic + accessible: Qure.ai (radiology imaging) https://qure.ai/ · SigTuple (microscopy/cell — closest to your fluorescence work) https://sigtuple.com/ · NVIDIA healthcare/MONAI + CUDA/TensorRT https://www.nvidia.com/en-us/about-nvidia/careers/ · GE HealthCare (Bengaluru R&D) https://www.gehealthcare.com/ · Microsoft Research India https://www.microsoft.com/en-us/research/lab/microsoft-research-india/ · Google Research India (stretch now, real by Tier 1) https://research.google/locations/india/

Tier 1 — broaden: Aidoc, Rad AI, Annalise.ai, PathAI, Paige.AI, Owkin (Paris/NYC) https://www.owkin.com/

Tier 2 — AI drug discovery + research labs (relocate): Recursion (*phenomics = deep learning on cell images — a direct match*) https://www.recursion.com/careers · Insitro https://insitro.com/ · Genesis Therapeutics https://www.genesistherapeutics.ai/ · Iambic https://www.iambic.ai/ · Chai Discovery https://www.chaidiscovery.com/ · EvolutionaryScale https://www.evolutionaryscale.ai/ · Profluent https://www.profluent.bio/ · InstaDeep https://www.instadeep.com/ · Valence Labs https://www.valencelabs.com/ · **Isomorphic Labs** https://www.isomorphiclabs.com/careers · **Google DeepMind** https://deepmind.google/about/careers/

- *Target the **Research Engineer / ML Research Engineer** postings* (verified to accept MSc + ~2 yrs ML experience — no PhD). Save Research Scientist for after a PhD.
- *Neutral nuance:* Isomorphic has a large drug-discovery partnership with Eli Lilly — not a blocker, but apply with your normal discretion.

QML career branch *(new in v8)* — quantum + AI for science (leverages Phase Q + your QpiAI history): QpiAI (Bengaluru; QpiAI-Pharma for drug discovery; quantum-chemistry / quantum-ML / AI-native roles — a natural fit given your vendor-oversight history) https://www.qpiai.tech/careers · IBM Quantum https://www.ibm.com/quantum · quantum-for-chemistry / materials teams broadly. *Pull Phase Q earlier only if this branch is a genuine target; otherwise QML stays a long-horizon investment + optionality.*

Tier 3 — top research loops: Isomorphic, Google DeepMind, Google Research (global + India), Genentech gRED / Prescient Design, Microsoft Research Health Futures, Meta FAIR.

Conflict-of-interest guardrails

Not legal advice — verify specifics with the right people at Lilly.

1. **Public data only.** Public benchmarks (Medical Decathlon/BraTS, QM9, PCQM4Mv2, CalMS21, PMC-OA, etc.). Never Lilly proprietary data, targets, molecules, assays, or internal docs. In Capstone Q/Capstone 3 use textbook/public molecules (H_2/LiH/H_2O/QM9), **not** your Lilly molecules (avoid the keto-enol / HATU work). **Also keep clear of anything specific to the QpiAI retrosynthesis engagement** you oversaw — public methods and public molecules only.
2. **Generic, published methods** in any portfolio artifact — no Lilly (or vendor) trade secrets or unpublished techniques.
3. **Personal time and resources** — your own hardware/accounts; not Lilly compute, licenses, or work hours. *(Any work on Lilly hardware/networks must use Lilly's JFrog Artifactory for packages per enterprise policy; these capstones are explicitly personal-hardware/personal-account, outside that environment.)*
4. **Clear before you publish** — run any paper/blog/repo/talk through Lilly's standard external-publication / IP-review process; when unsure, ask first.
5. **Check your agreement** — review employment / invention-assignment / moonlighting terms.
6. **Positioning stays generic** — transferable skills, never Lilly (or vendor) specifics.

Cross-cutting habits (every week, no exceptions)

De-duplicated: old habits #3, #6, #7 now live inside Threads M and R.

1. **Paper-of-the-week** — one paper, deeply read, derivations redone on paper. (Supersedes "1 paper/day.")
2. **Whiteboard Friday** — one concept whiteboarded without notes; photograph it → interview-prep deck.
3. **Sunday writeup** — a 1-page note teaching the week's deepest concept to a hypothetical junior.
4. **War chest + living glossary** — one doc for derivations, paper one-liners, gotchas, anti-patterns, and every confusing term in your own words (Confusion Buffer #2).

*(Re-implementation Saturday → **Thread M**. Reproduce-a-target-paper + positioning doc → **Thread R**.)*

PhD-later decision guide

Whether. For the long-term **Senior-RS** aspiration, a PhD is effectively required at the target labs (Isomorphic's RS posting asks for a PhD + publications; its RE posting accepts MSc + ~2 yrs, no PhD). So: RS title → PhD; senior/staff RE → PhD optional. *Note: your Phase-Q QML depth + the molecular spike also make you a credible candidate for a **quantum-ML / computational-chemistry PhD**, which is a distinct, well-funded niche if you want it.*

When. *After* landing a strong RE role — job-first expands options.

Where (and how job-first improves each):

- **India — part-time / external (moonlight-compatible).** IISc's **External Registration Programme** and the IITs' external/part-time PhD tracks admit working professionals (≈2 yrs' experience; IISc needs ~1 yr on-campus + a company co-advisor; part-time ~6 yrs). — https://www.iisc.ac.in/admissions/external-registration-programme-ph-d/
- **EU — salaried / industrial (a career move).** NL/Germany/Sweden/Switzerland/Austria PhDs are *salaried employee* roles (~€3–6k/mo, ~4 yrs); MSCA Industrial Doctoral Networks embed the PhD across a company + university. The strongest prestige-and-funding leap. — https://euraxess.ec.europa.eu/
- **USA — mostly full-time funded; part-time rare.** More realistic as a research-embedded path, or skip it via the RE→staff ladder.

Set-up now (zero marginal cost): the spike capstone → a non-archival workshop paper + a bioRxiv/arXiv preprint; 1–2 advisor relationships (reproduce a lab's work, engage at the workshop, email); Capstone Q gives you a *second* research identity (QML) if you want the quantum-PhD route.

Decision criteria (priority order): advisor fit → topic continuity → funding model (salaried/industrial > self-funded) → part-time feasibility → location/visa → prestige (last).

Publication & portfolio strategy

Realistic output for a working professional over ~18–24 months: 1–2 non-archival workshop papers + 1 preprint + clean open-source repos — not a solo top-conference main-track paper.

- **Job lever (near-term).** Reproduce-a-paper + a workshop paper = paper-discussion prep, referral hook, proof of level. MLSB prioritizes **open-sourced code** for spotlights — the code *is* the credential.
- **PhD-optionality lever.** First-author workshop papers + preprints + an advisor relationship lift you from "self-funded tier-2/3 PhD" to "funded EU/industrial or research-embedded."
- **Venues & timing.** Imaging/video → MIDL / MICCAI workshop (spring/summer) https://midl.io/ · https://www.miccai.org/ . Molecular → MLSB (NeurIPS, ~Sep/Oct submission, Dec) https://www.mlsb.io/ · LoG https://logconference.org/ · AI4Science / Simbiochem (NeurIPS). QML → a quantum-ML workshop or arXiv. Target **MIDL-2027 / MLSB-2027 + a rolling bioRxiv/arXiv preprint** you can cite immediately.
- **Visibility:** open-source every capstone with a reproducible README; contribute a small real fix to an ecosystem repo (**e3nn / MACE / PyG / MONAI / PennyLane**) — a merged PR beats another blog post.

Risk register

Risk	Likelihood	Impact	Leading indicator	Mitigation
Confusion compounding (carrying shaky foundations forward) — *your key risk as a slow-intuitive learner*	High	High	Self-tests failed or skipped; a term keeps reappearing unexplained	The **Confusion Buffer** — understanding-gated checkpoints, spaced re-derivation, consolidation weeks, multi-teacher triangulation. Do **not** advance until the Feynman gate passes
Burnout	High	High	Skipped rest/consolidation weeks; capstone quality dropping; dread	Steady ~25–35 hrs/wk, not heroic weeks; threads + consolidation weeks are shock absorbers; protect one rest day
Scope creep from Phase Q (QML pulling focus off the job path)	Med	High	Doing quantum before Tier-2 without a QML-role target	Keep Phase Q **off the critical path** (parallel-lite or post-Tier-2); pull earlier only for a QpiAI-type target
Breadth-without-depth	Med	High	Can *use* but can't derive/teach a topic	Self-tests are the gate; Thread M + Whiteboard Friday enforce depth; cut P7/legacy-P8 to protect the spike
Over-claiming CV seniority	Med	High	Positioning says "senior CV engineer"; freeze on classical-vision/SLAM	Reposition as "medical-imaging + geometric/molecular ML"; CV breadth = *working knowledge*
Goal drift / re-bloating	Med	Med	Adding "just one more" course	Every addition must serve imaging-video **or** molecular-geometric **or** the RE loop **or** (deliberately, off-path) QML — else cut
Relocation-vs-domestic split	Med	Med	Effort split 50/50, mediocre at both	Two legs share an umbrella so effort compounds; apply Tier-0 India *now* while building the Tier-2 spike
Premature / ill-fitting PhD	Low–Med	High	Applying to a self-funded tier-2/3 PhD before landing the job	PhD is post-job; job-first expands options

---

Resource index (quick lookup) — *all v6/v7 URLs preserved*

Brush-up (P0) — StatQuest https://www.youtube.com/@statquest/playlists · https://statquest.org/ · 3B1B NN https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi

Foundations math (P1) — 3B1B LA https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab · 3B1B Calc https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr · MIT 18.06 https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/ · MIT 18.065 https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/ · Stat 110 https://projects.iq.harvard.edu/stat110/youtube · MIT RES.6-012 https://ocw.mit.edu/courses/res-6-012-introduction-to-probability-spring-2018/ · Matrix Cookbook https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf · Matrix Calculus https://explained.ai/matrix-calculus/ · Karpathy Z2H https://www.youtube.com/playlist?list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ · hub https://karpathy.ai/zero-to-hero.html

Later-threaded math — Boyd https://www.youtube.com/playlist?list=PL3940DD956CDF0622 · https://web.stanford.edu/~boyd/cvxbook/ · Ruder https://www.ruder.io/optimizing-gradient-descent/ · MacKay https://www.inference.org.uk/itila/book.html · Group theory https://www.youtube.com/results?search_query=group+theory+for+physicists+lectures · GDL lectures https://geometricdeeplearning.com/lectures/

Medical imaging (P2) — MRI Q&A https://mriquestions.com/index.html · OHBM https://www.youtube.com/@ohbmonline/playlists · Callaghan https://www.youtube.com/playlist?list=PLqEMVgX2VgZcG_xR1QRdsuc-X_zEgMx-Z · iBiology https://www.youtube.com/playlist?list=PLF513KEDjY9YBNhwMfX6jMI3PlpdW9TMP · AI for Medical Diagnosis https://www.coursera.org/learn/ai-for-medical-diagnosis · DigitalSreeni https://www.youtube.com/@DigitalSreeni/playlists · code https://github.com/bnsreenu/python_for_microscopists · Conv arithmetic https://arxiv.org/abs/1603.07285 · https://github.com/vdumoulin/conv_arithmetic · MONAI https://github.com/Project-MONAI/tutorials · https://www.youtube.com/@projectmonai/videos · nnU-Net https://github.com/MIC-DKFZ/nnUNet · TorchIO https://torchio.readthedocs.io/ · Stanford AIMI https://aimi.stanford.edu/education/educational-resources · CS231n https://www.youtube.com/playlist?list=PL3FW7Lu3i5JvHM8ljYj-zLfQRF3EO8sYv · https://cs231n.github.io/ · Metrics Reloaded https://www.nature.com/articles/s41592-023-02151-z · Seg papers: U-Net https://arxiv.org/abs/1505.04597 · V-Net https://arxiv.org/abs/1606.04797 · Attention U-Net https://arxiv.org/abs/1804.03999 · nnU-Net https://www.nature.com/articles/s41592-020-01008-z · Loss survey https://arxiv.org/abs/2006.14822 · Generalized Dice https://arxiv.org/abs/1707.03237 · Focal Tversky https://arxiv.org/abs/1810.07842 · Boundary loss https://arxiv.org/abs/1812.07032 · ViT https://arxiv.org/abs/2010.11929 · TransUNet https://arxiv.org/abs/2102.04306 · Swin-UNet https://arxiv.org/abs/2105.05537 · UNETR https://arxiv.org/abs/2103.10504 · Swin UNETR https://arxiv.org/abs/2201.01266 · SAM https://arxiv.org/abs/2304.02643 · MedSAM https://www.nature.com/articles/s41467-024-44824-z · MedNeXt https://arxiv.org/abs/2303.09975 · **Datasets:** Medical Decathlon http://medicaldecathlon.com/ · BraTS https://www.synapse.org/brats · TCIA https://www.cancerimagingarchive.net/ · IXI https://brain-development.org/ixi-dataset/ · LIVECell https://github.com/sartorius-research/LIVECell · Sartorius Kaggle https://www.kaggle.com/competitions/sartorius-cell-instance-segmentation · BBBC https://bbbc.broadinstitute.org/

Deep learning core (P3) — CS25 https://web.stanford.edu/class/cs25/ · https://www.youtube.com/playlist?list=PLoROMvodv4rNiJRchCzutFw5ItR_Z27CM · CS336 https://stanford-cs336.github.io/spring2024/ · https://www.youtube.com/playlist?list=PLoROMvodv4rOY23Y0BoGoBGgQ1zmU_MT_ · CS236 https://www.youtube.com/playlist?list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8 · Lilian Weng https://lilianweng.github.io/ · Annotated Transformer http://nlp.seas.harvard.edu/annotated-transformer/ · HF NLP https://huggingface.co/learn/nlp-course · HF CV https://huggingface.co/learn/computer-vision-course · HF Diffusion https://huggingface.co/learn/diffusion-course · RoPE https://arxiv.org/abs/2104.09864 · RMSNorm https://arxiv.org/abs/1910.07467 · FlashAttention https://arxiv.org/abs/2205.14135 · GQA https://arxiv.org/abs/2305.13245 · SwiGLU https://arxiv.org/abs/2002.05202 · Switch/MoE https://arxiv.org/abs/2101.03961 · ConvNeXt https://arxiv.org/abs/2201.03545 · DINOv2 https://arxiv.org/abs/2304.07193 · MAE https://arxiv.org/abs/2111.06377 · DDPM https://arxiv.org/abs/2006.11239 · Score-based https://yang-song.net/blog/2021/score/ · Diffusion (blog) https://lilianweng.github.io/posts/2021-07-11-diffusion-models/

Quantum & QML (Phase Q) — IBM Quantum Learning https://learning.quantum.ibm.com/ · Watrous https://www.youtube.com/playlist?list=PLOFEBzvs-VvrXTMy5Y2IqmSaUjfnhvBHR · Basics https://learning.quantum.ibm.com/course/basics-of-quantum-information · Algorithms https://learning.quantum.ibm.com/course/fundamentals-of-quantum-algorithms · General Formulation https://learning.quantum.ibm.com/course/general-formulation-of-quantum-information · Error Correction https://learning.quantum.ibm.com/course/foundations-of-quantum-error-correction · Qiskit https://www.youtube.com/@qiskit/playlists · QDA blog https://www.ibm.com/quantum/blog/iql-migration · PennyLane codebook https://pennylane.ai/codebook · PennyLane QML https://pennylane.ai/topics/quantum-machine-learning · PennyLane chemistry https://pennylane.ai/qml/demos_quantum-chemistry · MIT 8.04 https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/ · Quantum Country https://quantum.country/qcvc · Schuller https://www.youtube.com/playlist?list=PLPH7f_7ZlzxTi6kS4vCmv4ZKm9u8g5yic · Péré https://github.com/Christophe-pere/QML-Course · Hjorth-Jensen https://github.com/CompPhysics/QuantumComputingMachineLearning · VQE https://www.nature.com/articles/ncomms5213 · ADAPT-VQE https://www.nature.com/articles/s41467-019-10988-2 · DMET-VQE https://arxiv.org/abs/2108.08987 · QAOA https://arxiv.org/abs/1411.4028 · Hadfield https://arxiv.org/abs/1709.03489 · QML=kernels https://arxiv.org/abs/2101.11020 · Barren plateaus https://www.nature.com/articles/s41467-018-07090-4 · Expressibility https://arxiv.org/abs/1905.10876 · Advantage-goal https://arxiv.org/abs/2203.01340 · PQC encodings https://arxiv.org/abs/2008.08605 · Pauli grouping https://arxiv.org/abs/1907.13117

Production / MLOps (P5) — CS329S https://stanford-cs329s.github.io/ · Made With ML https://madewithml.com/ · Full Stack DL https://fullstackdeeplearning.com/course/2022/ · Chip Huyen https://huyenchip.com/ml-interviews-book/ · Hamel Husain https://www.oreilly.com/radar/what-we-learned-from-a-year-of-building-with-llms-part-i/ · Eugene Yan patterns https://eugeneyan.com/writing/llm-patterns/ · Eugene Yan evals https://eugeneyan.com/writing/llm-evaluators/ · Anthropic contextual retrieval https://www.anthropic.com/news/contextual-retrieval · Anthropic Constitutional AI https://www.anthropic.com/news/constitutional-ai · LlamaIndex https://docs.llamaindex.ai/ · RAGAS https://docs.ragas.io/ · DSPy https://dspy.ai/ · Raschka https://magazine.sebastianraschka.com/ · HF multi-GPU https://huggingface.co/docs/transformers/en/perf_train_gpu_many · vLLM https://arxiv.org/abs/2309.06180 · ZeRO https://arxiv.org/abs/1910.02054

DSA & C++ (Thread T) — learncpp https://www.learncpp.com/ · The Cherno https://www.youtube.com/@TheCherno · cppreference https://en.cppreference.com/ · NeetCode https://neetcode.io/ · LeetCode https://leetcode.com/ · MIT 6.006 https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/ · Abdul Bari https://www.youtube.com/@abdul_bari

Implement-by-hand ML (Thread M) — d2l.ai https://d2l.ai/ · labml.ai https://nn.labml.ai/ · Annotated Transformer http://nlp.seas.harvard.edu/annotated-transformer/ · micrograd https://github.com/karpathy/micrograd · nanoGPT https://github.com/karpathy/nanoGPT · DeepMind RE round guide https://igotanoffer.com/en/advice/google-deepmind-research-engineer-interview · JAX https://jax.readthedocs.io/ · Equinox https://docs.kidger.site/equinox/ · Anki (spaced repetition) https://apps.ankiweb.net/

Research output (Thread R) — Papers with Code https://paperswithcode.com/ · MICCAI https://www.miccai.org/ · MIDL https://midl.io/ · MLSB https://www.mlsb.io/ · LoG https://logconference.org/ · ISCB/ISMB https://www.iscb.org/ · arXiv https://arxiv.org/ · bioRxiv https://www.biorxiv.org/ · OpenReview https://openreview.net/

GenAI & Agents (P7) — HF LLM course https://huggingface.co/learn/llm-course · HF Agents https://huggingface.co/learn/agents-course · HF MCP https://huggingface.co/learn/mcp-course · Anthropic Building Effective Agents https://www.anthropic.com/research/building-effective-agents · DeepLearning.AI short courses https://www.deeplearning.ai/short-courses/ · LangGraph https://langchain-ai.github.io/langgraph/ · LoRA https://arxiv.org/abs/2106.09685 · QLoRA https://arxiv.org/abs/2305.14314 · InstructGPT/RLHF https://arxiv.org/abs/2203.02155 · DPO https://arxiv.org/abs/2305.18290 · CLIP https://arxiv.org/abs/2103.00020 · LLaVA https://arxiv.org/abs/2304.08485 · Flamingo https://arxiv.org/abs/2204.14198 · **Corpora:** PMC-OA https://www.ncbi.nlm.nih.gov/pmc/tools/openftlist/ · PubMed E-utilities https://www.ncbi.nlm.nih.gov/books/NBK25501/ · Semantic Scholar API https://www.semanticscholar.org/product/api

Video & imaging (P2B) + optional CV literacy (P8) — First Principles of CV https://fpcv.cs.columbia.edu/ · FPCV Coursera https://www.coursera.org/specializations/firstprinciplesofcomputervision · Cyrill Stachniss https://www.youtube.com/@CyrillStachniss · Michigan EECS 498 https://web.eecs.umich.edu/~justincj/teaching/eecs498/ · playlist https://www.youtube.com/playlist?list=PL5-TkQAfAZFbzxjBHtzdVCWE0Zbhomg7r · NeRF https://arxiv.org/abs/2003.08934 · 3D Gaussian Splatting https://arxiv.org/abs/2308.04079 · OpenCV https://docs.opencv.org/ · PyImageSearch https://pyimagesearch.com/ · ONNX https://onnx.ai/ · TensorRT https://developer.nvidia.com/tensorrt · **Video:** VideoMAE https://arxiv.org/abs/2203.12602 · TimeSformer https://arxiv.org/abs/2102.05095 · ViViT https://arxiv.org/abs/2103.15691 · SlowFast https://arxiv.org/abs/1812.03982 · SAM 2 https://arxiv.org/abs/2408.00714 · https://github.com/facebookresearch/sam2 · RAFT https://arxiv.org/abs/2003.12039 · https://github.com/princeton-vl/RAFT · ByteTrack https://arxiv.org/abs/2110.06864 · https://github.com/ifzhang/ByteTrack · OC-SORT https://arxiv.org/abs/2203.14360 · https://github.com/noahcao/OC_SORT · BoT-SORT https://arxiv.org/abs/2206.14651 · https://github.com/NirAharon/BoT-SORT · HOTA https://arxiv.org/abs/2009.07736 · MOTChallenge https://motchallenge.net/ · DAVIS https://davischallenge.org/ · YouTube-VOS https://youtube-vos.org/ · **Life-sciences video:** DeepLabCut https://www.deeplabcut.org/ · SLEAP https://sleap.ai/ · Cellpose https://www.cellpose.org/ · TrackMate https://imagej.net/plugins/trackmate/ · Ultrack https://github.com/royerlab/ultrack · Cell Tracking Challenge http://celltrackingchallenge.net/ · CalMS21 https://data.caltech.edu/records/s0vdx-0k302 · **Deploy datasets:** KITTI https://www.cvlibs.net/datasets/kitti/ · TUM RGB-D https://cvg.cit.tum.de/data/datasets/rgbd-dataset · ETH3D https://www.eth3d.net/

Geometric / Molecular ML (P6 — the spike) — CS224W https://web.stanford.edu/class/cs224w/ · https://snap.stanford.edu/class/cs224w-2023/ · playlist https://www.youtube.com/playlist?list=PLoROMvodv4rPLKxIpqhjhPgdQy7imNkDn · Bronstein GDL https://geometricdeeplearning.com/lectures/ · proto-book https://arxiv.org/abs/2104.13478 · PyG https://pytorch-geometric.readthedocs.io/ · DGL https://www.dgl.ai/ · DeepChem https://deepchem.io/tutorials/ · TDC https://tdcommons.ai/ · SchNet https://arxiv.org/abs/1706.08566 · DimeNet++ https://arxiv.org/abs/2011.14115 · GemNet https://arxiv.org/abs/2106.08903 · MolFormer https://arxiv.org/abs/2106.09553 · Equivariant (Cohen&Welling) https://arxiv.org/abs/1602.07576 · EGNN https://arxiv.org/abs/2102.09844 · e3nn https://e3nn.org/ · https://github.com/e3nn/e3nn · e3nn tutorial https://blondegeek.github.io/e3nn_tutorial/ · NequIP https://github.com/mir-group/nequip · https://www.nature.com/articles/s41467-022-29939-5 · MACE https://github.com/ACEsuit/mace · https://arxiv.org/abs/2206.07697 · Allegro https://github.com/mir-group/allegro · Equiformer https://arxiv.org/abs/2206.11990 · curated list https://github.com/Eipgen/Neural-Network-Models-for-Chemistry · AlphaFold 2 https://www.nature.com/articles/s41586-021-03819-2 · ESM https://github.com/facebookresearch/esm · ESM Atlas https://esmatlas.com/ · Boltz https://github.com/jwohlwend/boltz · OpenFold https://github.com/aqlaboratory/openfold · **Datasets:** QM9 https://quantum-machine.org/datasets/ · PCQM4Mv2 https://ogb.stanford.edu/kddcup2021/pcqm4m/ · MoleculeNet https://moleculenet.org/ · OGB https://ogb.stanford.edu/

Interview (P9) — NeetCode https://neetcode.io/ · LeetCode https://leetcode.com/ · DesignGurus https://www.designgurus.io/ · ByteByteGo https://bytebytego.com/ · https://www.youtube.com/@ByteByteGo · Gaurav Sen https://www.youtube.com/@gkcs · MIT 6.824 (optional) https://pdos.csail.mit.edu/6.824/

Company ladder — Qure.ai https://qure.ai/ · SigTuple https://sigtuple.com/ · NVIDIA https://www.nvidia.com/en-us/about-nvidia/careers/ · GE HealthCare https://www.gehealthcare.com/ · MSR India https://www.microsoft.com/en-us/research/lab/microsoft-research-india/ · Google Research India https://research.google/locations/india/ · Owkin https://www.owkin.com/ · Recursion https://www.recursion.com/careers · Insitro https://insitro.com/ · Genesis https://www.genesistherapeutics.ai/ · Iambic https://www.iambic.ai/ · Chai https://www.chaidiscovery.com/ · EvolutionaryScale https://www.evolutionaryscale.ai/ · Profluent https://www.profluent.bio/ · InstaDeep https://www.instadeep.com/ · Valence Labs https://www.valencelabs.com/ · Isomorphic Labs https://www.isomorphiclabs.com/careers · Google DeepMind https://deepmind.google/about/careers/ · **QML branch:** QpiAI https://www.qpiai.tech/careers · IBM Quantum https://www.ibm.com/quantum · **PhD routes:** IISc ERP https://www.iisc.ac.in/admissions/external-registration-programme-ph-d/ · EURAXESS https://euraxess.ec.europa.eu/

A note on what you're taking on

This is a **~1,870-hour** program on top of a full-time role — robust, honestly scoped, and built for *how you learn*: intuition first, generous time on the math and fundamentals where your real ceiling is, and a **Confusion Buffer** that turns confusing terminology from a wall into a queue. Anchors: the self-tests and Feynman gates are the real gates, not the calendar — **advance on understanding, not dates**; take consolidation weeks liberally; the threads are shock absorbers you dial down on heavy weeks. **Apply from Tier 0 now** — your applied imaging skill already makes you eligible, and early loops pay you in feedback while the foundations build underneath. Give the **QML deep track (Phase Q)** the real hours it deserves *because you want that expertise* — but keep it off the early-job path so it never delays your ladder; let it pay off later, in the molecular spike's depth, a QML career branch (QpiAI and beyond), and optional PhD routes. Re-activating and *building* the fundamentals will feel slow at first — that slowness is the point; it's what turns "I use this" into "I understand this well enough to push a model past its ceiling and teach it to a panel." Spend your best focus on the **foundations and the spike**, not the legacy — and let the incremental ladder carry you from your first India loop to a real shot at a life-sciences research-engineering seat, with a genuine QML capability and a PhD option waiting whenever you want them.